MA 324

Statistical Inference and Multivariate Analysis

IIT GUWAHATI

End-Semester Examination

14:00–17:00 IST

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PART-A

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Instructions

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	NA	04	7	12

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3,$ be the observed values of the response variable. Consider,

$$\begin{aligned} Y_1 &= \theta_1 + \theta_2 + \epsilon_1, \\ Y_2 &= \theta_1 - 2\theta_2 + \epsilon_2, \\ Y_3 &= 2\theta_1 - \theta_2 + \epsilon_3, \end{aligned}$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2.

2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\boldsymbol{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.
   (a) (3 points) Compute the correlation matrix $\boldsymbol{\rho}$ of $\boldsymbol{X}$.
   (b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

(a) computing the correlation matrix ρ of X

Prilimminary statistics for U_1 & U_2:

~~two independent uniform random variables (0,1) interval~~

~~Suppose $X = (X_1, X_2, X_3)$~~

for a uniform distribution $U \sim unif(0,1)$

$$E[U] = \frac{1}{2}$$

$$Var(U) = \sigma^2 = \frac{(1-0)^2}{12} = \frac{1}{12}$$

Since U_1, U_2 are independent:

$cov(U_1, U_2) = 0$

Variance co-variance matrix Σ:

we need $var(X_i)$ and $co\text{-}var(X_i, X_j)$ for all

$i, j \in \{1, 2, 3\}$

$$Var(X_1) = Var(U_1) = \frac{1}{12}$$

$$Var(X_2) = Var(U_2) = \frac{1}{12}$$

$$Var(X_3) = Var(U_1 + U_2) = Var(U_1) + Var(U_2)$$

by independence

$cov(X_1, X_2) = 0$ (given independence)

$cov(X_1, X_3) = Cov(U_1, U_1 + U_2) = Var(U_1) + cov(U_1, U_2) = \frac{1}{12} + 0 = \frac{1}{12}$

correlation coefficient ρ_{ij}:

The formula is:

$$\rho_{ij} = \frac{cov(X_i, X_j)}{\sqrt{Var(X_i) Var(X_j)}}$$

$\rho_{12} = 0$

$$\rho_{13} = \frac{1/12}{\sqrt{(1/12)(2/12)}} = \frac{1}{\sqrt{2}}$$

Final correlation matrix ρ

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$$\rho = \begin{bmatrix} 1 & 0 & 1/\sqrt{2} \\ 0 & 1 & 1/\sqrt{2} \\ \frac{1}{\sqrt{2}} & 1/\sqrt{2} & 1 \end{bmatrix}$$

b) characteristic eqn

~~det~~

$\det(\rho - \lambda I) = 0$

$$\begin{vmatrix} 1-\lambda & 0 & 1/\sqrt{2} \\ 0 & 1-\lambda & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1-\lambda \end{vmatrix} = 0$$

How ??

$\lambda = 2$

The eigen values are $\lambda_1 = 2, \lambda_2 = 1, \lambda_3 = 0$

The variance of the first principal component is the maximum eigen value.

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3. Let the matrix of distance be

$$
\begin{array}{c|cccc}
 & 1 & 2 & 3 & 4 \\
1 & \multicolumn{4}{l}{\left[\begin{array}{cccc} 0 & & & \\ 1 & 0 & & \\ 11 & 2 & 0 & \\ 5 & 3 & 3.5 & 0 \end{array}\right]} \\
2 & & & & \\
3 & & & & \\
4 & & & &
\end{array}
$$

(a) (6 points) Cluster the four items using each of the following procedures:

i. Complete linkage hierarchical procedure.

ii. Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

3(a)(i) The smallest distance $d(1,2)$ is 1.

merge $\{1\}$ & $\{2\}$ into cluster $\{1,2\}$

update distances:

$d(\{1,2\},3) = \max(d\{1,3\}, d\{2,3\}) = \max(1,2) = 11$

$d(\{1,2\},4) = \max(d\{1,4\}, d\{2,4\}) = \max(5,3) = 5$

new matrix

$$
\begin{array}{c|ccc}
 & \{1,2\} & 3 & 4 \\
\hline
\{1,2\} & 0 & \square & \square \\
3 & 11 & 0 & \square \\
4 & 5 & 3.5 & 0
\end{array}
$$

The smallest distance is $d(3,4) = 3.5$

merge $\{3\}$ & $\{4\}$ in cluster $\{3,4\}$

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update distances.

$d(\{1,2\},\{3,4\}) = \max(d(1,3), d(1,4), d(2,3), d(2,4)) =$
$\max(1, 5, 2, 3) = 11$

merge $\{1,2\}$, $\{3,4\}$ at distance 11

ii) $d(U,V) = \frac{1}{|U||V|} \sum_{u \in U} \sum_{v \in V} d(u,v)$

Smallest distance is $d(1,2) = 1$

Merge $\{1,2\}$

Update distances

$d(\{1,2\},3) = \frac{d(1,3) + d(2,3)}{2} = \frac{11+2}{2} = 6.5$

$d(\{1,2\},4) = \frac{d(1,4) + d(2,4)}{2} = \frac{5+3}{2} = 4$

New matrix

$$\begin{array}{c|cccc} & \{1,2\} & 3 & 4 \\ \hline \{1,2\} & 0 & \square & \square \\ 3 & 6.5 & 0 & \square \\ 4 & 4 & 3.5 & 0 \end{array}$$

the smallest distance is $d(3,4)$

update distances:

~~$d(\{1,2\},\{3,4\}) = d(1,3) + d(1,4) = \frac{11+5+2}{4}$~~

$d(\{1,2\},\{3,4\}) = \frac{d(1,3) + d(1,4) + d(2,3) + d(2,4)}{4} = \frac{11+5+2+3}{4}$

$= \frac{21}{4}$

Last step

merge $\{1,2\}$, $\{3,4\}$ at distance 5.25

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Part B on next Page

ADDITIONAL PAGES FOR ANSWERS

b) complete clendroit linkage

cluster {1,2} at heigh 1

Dendogram 2?

cluster {3,4} at height 3.5

joins the 2 main cluster at height 11.

Average linkage clendograms.

cluster {1,2} at heigh 1

cluster {3,4} at height 3.5

joinds the 2 main cluster at height 5.25

Comparizon:

Both methods yield the same cluster membership at each step. First {1,2} then {3,4} and finally everything together.

However heights differ significantly. Complete distance is sensitive to outerdistance between 1 & 3 (d=11), forcing the final merge to happen at a much higher threshold compared to average linkage, which smooths out that ~~average distance~~ extreme distance

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